

AES Model with Logic Rule Reasoning and Self-Explanation Based on AMR and LLM

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Abstract—Automated Essay Scoring (AES) has great potential in real-world applications. However, AES models function as black boxes, which makes it challenging for users to comprehend the rationale behind scoring decisions and ultimately diminishes trust. Second-language learners, in particular, need explanations to improve their writing skills. Currently, there are no explainable AES methods in deep learning. In this paper, we introduce Large Language Models (LLMs) and Abstract Meaning Representation (AMR) techniques and integrate existing self-explanation frameworks to create the first interpretable AES model. We use LLMs to generate detailed explanations and select high-quality ones for rule generation. AMR technology then breaks these explanations into generalizable logical rules. We integrate explanation generation and score prediction in an end-to-end manner, optimizing through joint learning. Experimental results indicate that our method surpasses prompt-tuned ChatGPT-4, providing logical rule-based explanations that align with human cognitive habits and reducing costs. Our approach combines the knowledge generation capability of large models with the supervised learning abilities of smaller models.

Keywords—Automated Essay Scoring; Self-explaining; Logic rule; Large Language Models; Abstract Meaning Representation

I. INTRODUCTION

Deep learning has delivered outstanding performance across numerous natural language processing tasks, often surpassing human performance in areas such as machine translation, text summarization, and question answering [1], [2]. However, due to the complex and opaque nature of deep networks, they are often criticized as black-box models, raising concerns about their transparency, ethics, and trustworthiness. This opacity hinders verifying prediction reliability. In actually, both weak and strong models benefit from interpretability: weaker models need it to correct errors, while stronger models, like AlphaGo, could use it to help

humans make better decisions. Despite substantial progress in automatic essay scoring, limited research has been done on its interpretability. In China, where the teacher-student ratio is high, providing personalized feedback on English writing is a challenge. Achieving interpretability in essay scoring has practical and educational value in addressing this issue. For example, providing personalized feedback through an explainable model can quickly improve students' writing skills. Xiao et al. [3] fine-tuned ChatGPT-4 for essay explanation and scoring using prompt-tuning techniques. While this approach offers valuable feedback to improve efficiency, its high costs make it unsustainable. Lee et al. used a self-explanatory framework with logical rules to generate scores and feedback, but their use of individual words as atomic rules limits semantic richness. We build on Lee et al.'s self-explanatory model [4] and optimize it further. Recent interpretability research focuses on post-hoc explanations for black-box models, which maintain predictive performance but face challenges like high computational costs and trust issues [4], requiring extensive tests to fully explain model behavior [5]. Self-explaining methods effectively avoid issues by integrating the explanation mechanism into the model, allowing predictions and explanations to occur simultaneously without relying on approximations that may undermine credibility. However, these methods struggle to ensure that humans can naturally deduce the correct answer from the explanations. For example, the explanation "happy $\geq$ 2" intuitively suggests a positive sentiment, while "is $\geq$ 2" does not. To address this, Lee et al. [4] chose logical rules as the minimal atoms of explanation and optimized rule confidence to generate explanations. But, in our task, simple word-based feedback is insufficient for improving writing skills, raising the need for better logical rule selection in self-explaining essay scoring. To tackle

these issues, we introduced the open-source large model ChatGLM3 to generate personalized essay comments based on Lee’s work, which will serve as the basis for the rules. We also upgraded the rules from words to triples <subject, predicate, object>, aligning with human cognitive habits and effectively explaining essays. Specifically, we use AMR technology to convert comments into AMR graphs and extract triples from these graphs as logical rules.

In summary, our specific contributions are as follows:

- Our work introduces the first self-explanatory framework for automatic essay scoring, offering feedback based on logical rules alongside scoring, advancing the market application of AES models.
- We utilized the open-source ChatGLM3 model for low-cost personalized explanatory comments via prompt tuning, supplying raw material for generating logical rules and enriching future research on explainable essay scoring.
- We enhanced the semantically simple word logical rules to richer, more explanatory tuple logical rules using AMR technology. Explanations composed of semantically rich tuple logical rules align better with human thinking patterns.

II. RELATED WORK

This work mainly involves two aspects of research: Automatic Essay Scoring (AES) and the interpretability of natural language processing tasks.

A. Automated Essay Scoring

Research on automatic essay scoring (AES) has been ongoing for 50 years [6]. From the perspective of how essays are represented in models, existing AES methods are classified into early handcrafted feature-driven approaches [7], recent deep learning-based methods [8], and mixed feature-based methods [9]. These three approaches can be further summarized into three scoring modes: prompt-specific [10], prompt-adaptive [11], and prompt-generalized [12]. Although prompt-specific methods can achieve better scoring performance, prompt-adaptive and prompt-generalized methods can effectively reduce the annotation workload in target prompts.

B. Explainability in Deep Learning

Recent advances in deep learning have drawn attention due to their exceptional performance, but concerns over their black-box nature have prompted research on interpretability. Marco et al. [5] introduced LIME to explain model predictions through locally interpretable models. Rajani et al. [13] collected commonsense reasoning explanations with the CoS-E dataset but faced replication challenges due to complex annotations. To avoid this, Lee et al. [4] proposed a self-explaining model using logical rules, but their approach lacks semantic depth for tasks like essay scoring. Our

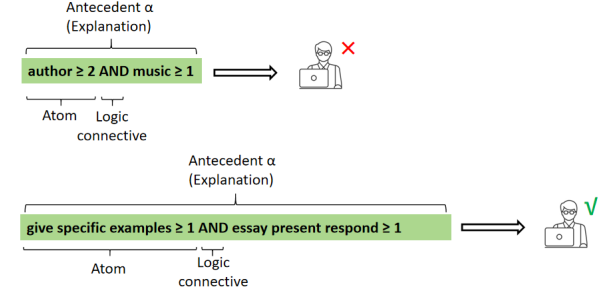


Figure 1. Compared to individual words, triplet phrases have stronger explanatory power.

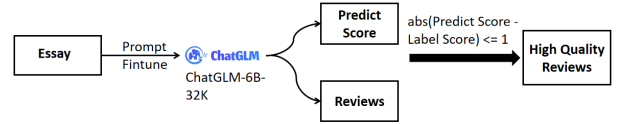


Figure 2. Prompt-tune ChatGLM and extract high-quality essay reviews.

work builds on Lee et al. [4], using ChatGLM3 and AMR technology to generate semantically rich triples for essay feedback, introducing the first self-explanatory model for automatic essay scoring.

III. METHOD

A. Formulation of Logic Rules

Our task uses logical rules $\alpha \Rightarrow y$ with antecedent α (prerequisite or explanation) and consequent y (predicted output). α consists of atoms or logical connectives, where atoms are smallest explanatory units (words or sentences). Each atom o_i is a Boolean function based on input presence. Logical connectives can be common or custom. To explain essay scoring, we use ChatGLM3 with prompt tuning for personalized feedback, then apply AMR to extract triples as antecedent α . As shown in Fig. 1, this outperforms using individual words as antecedent.

B. Generation of Logical Rules

1) *Prompt-tuning ChatGLM3*: Due to the need for external knowledge to provide explanations for automated essay scoring, we have selected the latest version of the ChatGLM series, ChatGLM3, an open-source large language model jointly developed by Tsinghua University and Zhipu AI, as the generator of personalized explanatory feedback for essays. Following the work of Xiao et al. [3], we crafted prompts and fine-tuned ChatGLM to simultaneously generate essay scores and personalized feedback. To ensure the authenticity of the feedback, we preserved high-quality feedback where the absolute gap between the predicted and label score is no greater than 1. The process of fine-tuning ChatGLM3 to generate feedback is shown in Fig. 2.

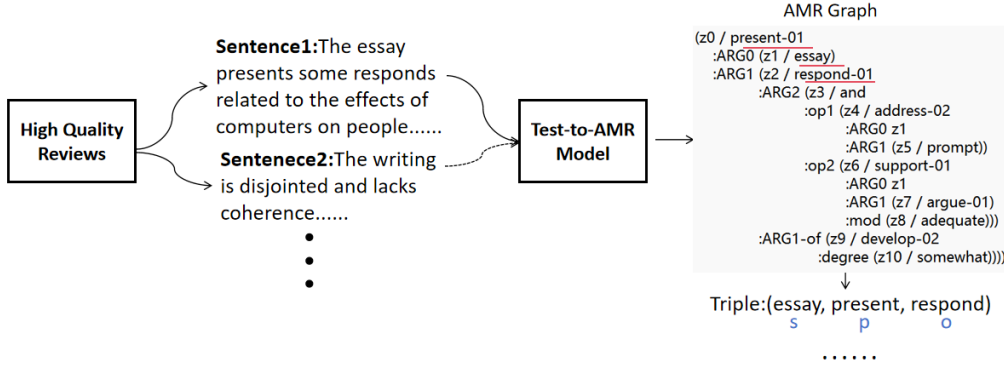


Figure 3. We use the Test-to-AMR model to extract the core semantic triples from high-quality essay feedback. In the Figure, s represents the subject, p represents the predicate, and o represents the object.

2) *From AMR to Triples*: Since each essay's comments are different, using them as explanatory atoms lacks generalizability. Moreover, it is evident that even individual words from the comments cannot effectively explain the essay. Considering that AMR (Abstract Meaning Representation) can transform sentences from natural language into structured semantic representations, capturing the core meaning and role relationships of the sentences, it helps the model understand the deeper meaning of the language. We use the end-to-end Text-to-AMR parsing framework proposed by Bevilacqua et al. [14] to perform AMR parsing on each sentence in the comments. Furthermore, by analyzing the AMR graph, we can extract subject-verb-object triples from the sentences in the comments. The extraction process of the triples is shown in Fig. 3.

C. Framework for Deep Logic Rule Reasoning

We define f as a deep learning model predicting $p(y|\mathbf{x})$, where $\mathbf{x}$ is the input and y is the candidate class. To make f self-explanatory, we introduce a latent variable α as a logical explanation, and redefine f as:

$$p(y|\mathbf{x}, b) = \sum_{\alpha} p(y|\alpha, \mathbf{x}, b) p(\alpha|\mathbf{x}, b) = \sum_{\alpha} p(y|\alpha) p(\alpha|\mathbf{x}, b), \text{ s.t., } \Omega(\alpha) \leq S \quad (1)$$

Here, b represents human priors, specifying the types of rules (e.g., atoms, logical connectives). $\Omega(\alpha)$ denotes the rule complexity, bounded by S to ensure readability. In our case, $S = 1$. Eq. 1 imposes two constraints: one ensuring, $p(y|\alpha, \mathbf{x}, b) = p(y|\alpha)$ so that α incorporates $\mathbf{x}$ and b ; and the second limits the number of explanations. Expanding this, we get:

$$p(y|\mathbf{x}, b) = \sum_{\alpha} p(y|\alpha) p(\alpha|\mathbf{x}, b) \propto \sum_{\alpha} p(b|\alpha) \cdot (y|\alpha) \cdot (\alpha|\mathbf{x}), \text{ s.t., } \Omega(\alpha) \leq S \quad (2)$$

The three components in Eq. 2 form the core framework of our model, with the detailed structure illustrated in Fig. 4.

- **Human prior** $p(b|\alpha)$ sets the format of the rules according to the ideal model of human understanding to ensure the readability of the rules.
- **Consequent estimation** $p(y|\alpha)$ ensures the necessary connection between the explanation and the predicted result, allowing humans to naturally associate the prediction with the explanation.
- **Deep antecedent generation** $p(\alpha|\mathbf{x})$ ensures that the deep model f can generate explanation α from the deeper meaning of the input $\mathbf{x}$.

IV. EXPERIMENT

Settings We conducted our experiments on the ASAP dataset¹, a publicly available English dataset for the automated essay scoring task, funded by the William and Flora Hewlett Foundation. We used Cohen's [15] quadratic weighted kappa for evaluation, which accounts for the possibility of random classification and gives more weight to matches while penalizing mismatches more than linear weighting. In our experiments, the batch size was set to 4, the number of training epochs to 20, the learning rate to $1e-5$, the number of atom to 5000, the antecedent length to 4, and "and" was selected as the logical connective. We selected the ChatGLM model version ChatGLM3-6B-32K².

Baselines We selected two baseline methods, as shown in Table I: Lee [4] and Xiao et al. [3]. Lee [4] uses words

¹The dataset is available at <https://www.kaggle.com/c/asap-aes/data>

²<https://github.com/THUDM/ChatGLM3>

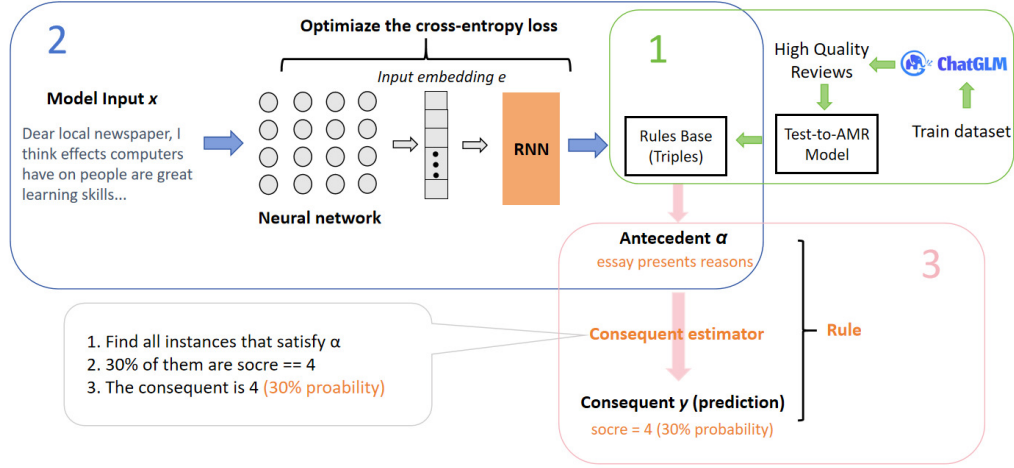


Figure 4. The overall flowchart of our method. Part 1 represents the construction process of the rule base, Part 2 represents the mapping process from input to rules, and Part 3 represents the mapping process from rules to results.

Table I
SELOR AND CHATGPT-4 ARE THE TWO BASELINE METHODS. P1 REPRESENTS PROMPT1, AND SO ON. WE USE THE BOLDED FONT TO INDICATE THE OPTIMAL EFFECT AND USE KAPPA VALUE AS THE EVALUATION METRIC

Method	P1	P2	P3	P4	P5	P6	P7	P8
SELOR [4]	0.011	0.013	0.012	0.025	0.022	0.024	0.01	0.014
ChatGPT-4 [3]	0.0715	0.3003	0.3661	0.6266	0.5227	0.3448	0.1101	0.4072
Our Method	0.1205	0.346	0.3915	0.5391	0.628	0.3866	0.1416	0.3374

Table II
SELOR REPRESENTS THE ORIGINAL METHOD, SELOR_{LLM} REPRESENTS THE INCORPORATION OF CHATGLM3 ON THE BASIS OF THE ORIGINAL METHOD, AND SELOR_{LLMandAMR} REPRESENTS THE INTEGRATION OF BOTH CHATGLM3 AND AMR TECHNOLOGY ON THE BASIS OF THE ORIGINAL METHOD

Method	P1	P2	P3	P4	P5	P6	P7	P8
SELOR [4]	0.011	0.013	0.012	0.025	0.022	0.024	0.01	0.014
SELOR _{LLM}	0.1005	0.3887	0.3221	0.5127	0.5936	0.35407	0.1128	0.3254
SELOR _{LLMandAMR}	0.1205	0.346	0.3915	0.5391	0.628	0.3866	0.1416	0.3374

as explanatory atoms, lacking semantics, resulting in low prediction accuracy and poor explanatory performance of the model. Xiao et al. [3] improves explanatory performance, but its understanding of long texts is slightly poorer due to limitations imposed by prompt fine-tuning.

Our Method We made improvements based on Lee [4]’s method by using triples as explanation atoms, which not only ensured the generalization of the model but also enhanced both prediction accuracy and the readability of the explanations. We trained and tested the model within the data of each prompt separately. The results, presented in Table I, show that our method surpasses the baseline model under most prompts, demonstrating its effectiveness and indicating that the model’s explanations align with human comprehension habits.

Ablation Experiments We conducted three ablation experiments to explore the role of each component. Firstly, we experimented using the original method, SELOR [4]. Then, we added ChatGLM3 on the basis of SELOR [4]

for further experimentation. Finally, we conducted experiments by incorporating both ChatGLM3 and AMR into SELOR [4], which is also the method we proposed. The experimental results, as shown in Table II indicate that our method performs optimally across 7 prompts. Moreover, both SELOR_{LLM} and SELOR_{LLMandAMR} significantly outperform the original SELOR [4], demonstrating that ChatGLM3 can provide prior explanations for small models, and that AMR technology further enables the generalization of rules.

V. CONCLUSION

Our approach proposed the first explainable AES model in NLP, bridging a gap. ASAP dataset experiments showed its superior prediction to ChatGPT-4 with high explainability. We improved Lee et al [4]’s method by upgrading rules to triple-level via LLM fine-tuning and AMR, enhancing explainability. Future plans include elevating triple-level rules to sentence-level with retrieval-augmented techniques,

aligning with human reading habits. Given kappa value room for improvement, we'll build datasets to fine-tune ChatGLM for better essay commenting and prediction accuracy.

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